

Exp No: 13

Code:

```
#include <reg51.h>
```

```
void main(void)
```

```
{
```

```
    unsigned int i;
```

```
    unsigned int array[5] = {0x1111, 0x2222, 0x8888, 0x4444, 0xABCD};
```

```
    unsigned long sum = 0;
```

```
    for (i = 0; i < 5; i++)
```

```
    {
```

```
        sum = sum + array[i];
```

```
    }
```

```
    P0 = (unsigned char)(sum & 0xFF);
```

```
    P1 = (unsigned char)((sum >> 8) & 0xFF);
```

```
    P2 = (unsigned char)((sum >> 16) & 0xFF);
```

```
    while (1);
```

```
}
```

Output:

The screenshot displays the uVision IDE interface. The main window shows the disassembly and source code of a program. The source code is as follows:

```
14: P0 = (unsigned char)(sum & 0xFF);  
15: MOV A, 0x15  
16: CLR A  
17: for (i = 0; i < 5; i++)  
18: {  
19:     sum = sum + array[i];  
20: }  
21: P0 = (unsigned char)(sum & 0xFF);  
22: P1 = (unsigned char)((sum >> 8) & 0xFF);  
23: P2 = (unsigned char)((sum >> 16) & 0xFF);  
24: while(1);
```

The Watch window shows the current state of variables:

Name	Value	Type
0x00	0x0000	int
sum	0x0001ABCC	ulong
array[0]	0x1111	uint
array[1]	0x2222	uint
array[2]	0x8888	uint
array[3]	0x4444	uint
array[4]	0xABCD	uint
array[5]	0x0001	uint
Address: 0x00	<cannot evaluate>	uchar
Address: 0x00	<cannot evaluate>	uchar